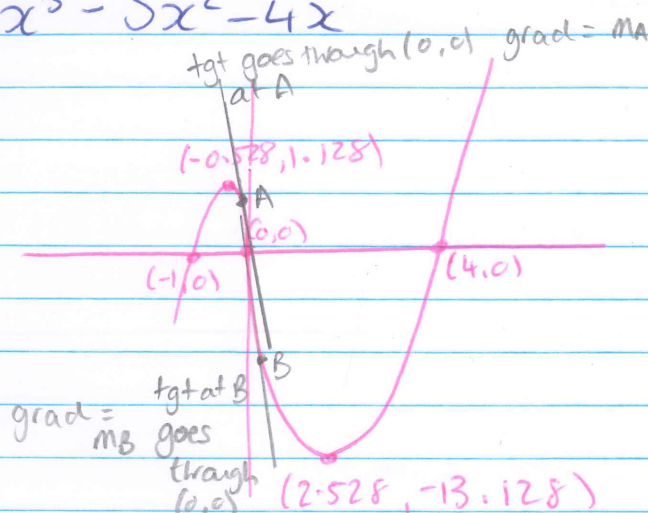


Q3c

$$f(x) = x^3 - 3x^2 - 4x$$



graphics calculator.

$$f'(x) = 3x^2 - 6x - 4$$

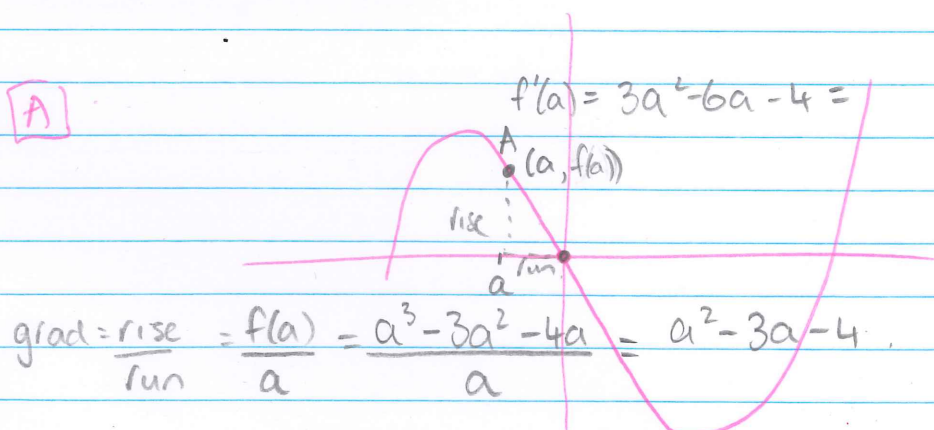
for point A - let gradient at A be m_A
and let x-value at A be 'a'

then $f'(a) = 3(a)^2 - 6(a) - 4 = m_A$
and as tangent A goes through (0,0)

$$y_A = (3a^2 - 6a - 4)x$$

Sub in (0,0) won't work $\rightarrow$ Arrgh!

A



$$f'(a) = 3a^2 - 6a - 4 = a^2 - 3a - 4$$

$$2a^2 - 3a = 0$$

$$a(2a - 3) = 0$$

$$a = 0 \text{ or } a = \frac{3}{2}$$

when $a = 0$

co-ordinate is $(0, 0)$

$$f'(0) = -4$$

$$\text{tgt } y = -4x$$

when $a = \frac{3}{2}$

$$f(a) = \left(\frac{3}{2}\right)^3 - 3\left(\frac{3}{2}\right)^2 - 4\left(\frac{3}{2}\right)$$

$$= -\frac{75}{8} = -9.375$$

$$f'\left(\frac{3}{2}\right) = -\frac{25}{4}$$

co-ord is $\left(\frac{3}{2}, -\frac{75}{8}\right)$

$$\text{tgt } y = -\frac{25x}{4}$$

Grades.

ach: finds $f'(x)$ and makes some progress

merit: finds one point and equation of the tangent

exc: finds both points and equations of both tangents.